

ALA: 1-Python

AIM: To write a Python program that generates the terms of a Geometric Progression using a while loop, checks whether each term is even or odd, calculates the total sum of the terms, and determines whether the series is constant or changing.

Error code:

Original Code (With Errors)

```
print("Geometric Progression Tracker")
a = int(input("Enter first term: "))
r = int(input("Enter ratio: "))
n = int(input("Enter terms: "))
i = 1
sum_total = 0
while i <= n:
    term = a * (r ** i)
    sum_total = sum_total + term
    if term % 2 == 0:
        print("Even term:", term)
    else:
        print("Odd term:", term)
    i = i + 1
print("Total sum:", sum_total)
```

```
if r = 1:  
    print("Constant series")  
else:  
    print("Changing series")
```

Error : 1- Indentation Error

- Python uses **indentation (spaces)** to define blocks of code.
Inside **while** and **if**, your lines were **not indented**.

- Wrong :

```
while i <= n: term = a * (r ** i)
```

- Correct :

```
while i <= n: term = a * (r ** i)
```

- **ERROR 2: Wrong GP Formula (Logical Error)**

- Problem :

```
term = a * (r ** i)
```

- Fixed Code :

```
term = a * (r ** (i - 1))
```

- **ERROR 3: Indentation inside if-else**

- Problem:

```
if and else blocks were not indented
```

- Wrong :

```
if term % 2 == 0:  
    print("Even term:", term)
```

- Correct :

```
if term % 2 == 0:  
    print("Even term:", term)  
else:  
    print("Odd term:", term)
```

- **ERROR 4: Missing Closing Parenthesis**

- Problem :

```
print("Total sum:", sum_total
```

- Correct :

```
print("Total sum:", sum_total)
```

- **ERROR 5 : Using = instead of ==**

- Problem :

```
if r = 1:
```

- Correct :

```
if r == 1:
```

- **ERROR 6: Wrong Placement of if-else (Logical Error)**

- **Problem :**

The series type check was placed **inside the loop logically**, but it should run **once after the loop**.

- **Correct Placement :**

```
print("Total sum:", sum_total)
```

```
if r == 1:
```

```
    print("Constant series")
```

```
else:
```

```
    print("Changing series")
```

- **ERROR 7: No Proper Block Structure**

- **Problem :**

- **Code mixed:**

- loop code
- print statements
- condition checks

Solution:

```
print("Geometric Progression Tracker")
```

```
a = int(input("Enter first term: "))
```

```
r = int(input("Enter ratio: "))
```

```
n = int(input("Enter terms: "))
```

```
i = 1
```

```
sum_total = 0
```

```
while i <= n:
```

```
    term = a * (r ** (i - 1))
```

```
    sum_total = sum_total + term
```

```
    if term % 2 == 0:
```

```
        print("Even term:", term)
```

```
    else:
```

```
        print("Odd term:", term)
```

```
    i = i + 1
```

```
print("Total sum:", sum_total)
```

```
if r == 1:
```

```
    print("Constant series")
```

```
else:
```

```
    print("Changing series")
```

OUTPUT:

```
C:\Users\admin\PycharmProjects\dhvani\.venv\Scripts\python.exe C:\Users\admin\PycharmProjects\dhvani\ala.py
Geometric Progression Tracker
Enter first term: 2
Enter ratio: 2
Enter terms: 4
Even term: 2
Even term: 4
Even term: 8
Even term: 16
Total sum: 30
Changing series
```